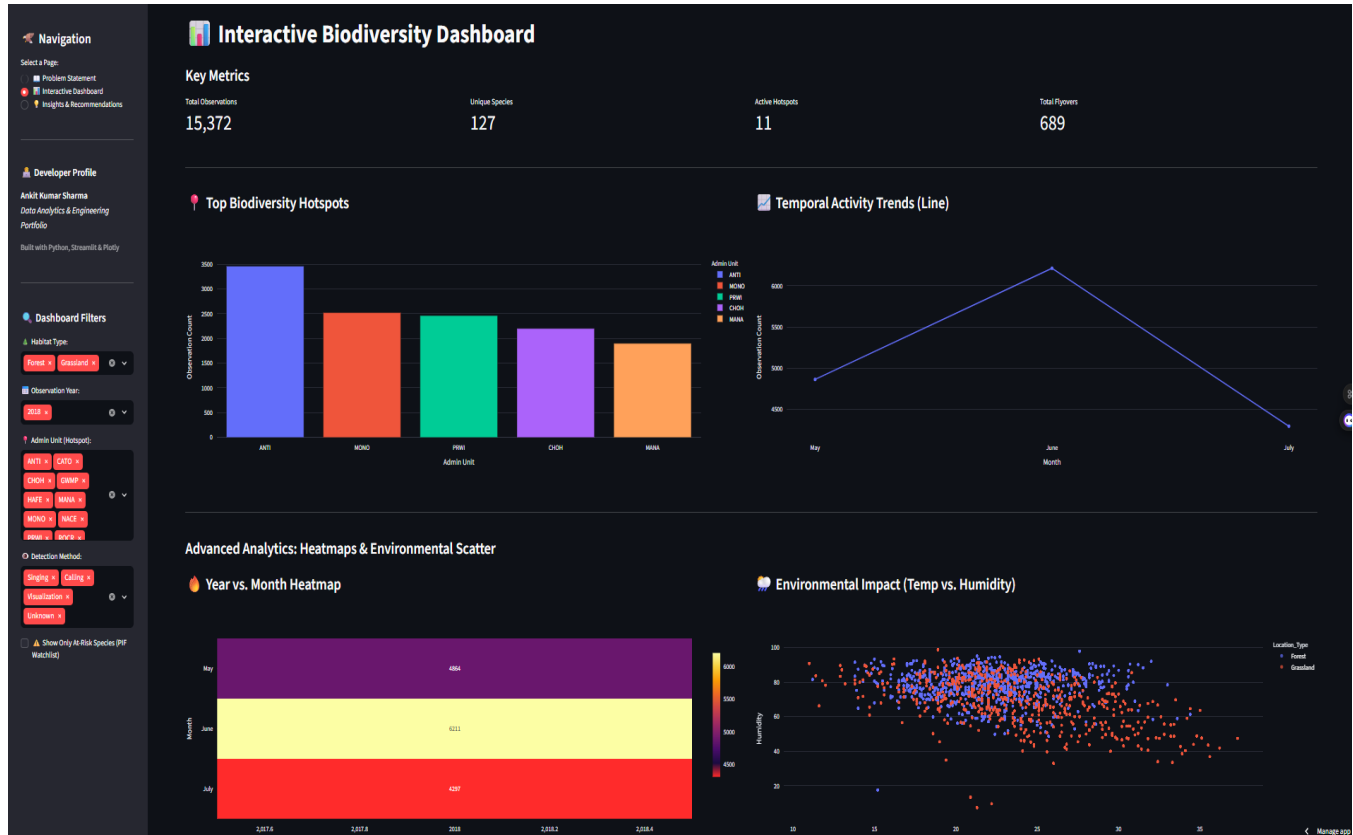


Bird Species Observation Analysis

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Final Dashboard View



Stage 0: Problem Statement & Objective

- **Project Goal:** The objective of this project is to transform raw, fragmented ecological data into a centralized, interactive intelligence tool. The primary stakeholders are environmental managers, conservationists, and eco-tourism directors who need to spot patterns in habitat preferences and species behavior to make data-driven decisions.
- **Key KPIs Tracked:** Total Observations, Unique Species, Active Hotspots, Total Flyovers, and At-Risk Species counts.
- **Data Preparation:** The project was executed in four distinct technical phases: 1) Data Engineering & ETL, 2) Relational Database Architecture, 3) Exploratory Data Analysis (EDA), and 4) Interactive Dashboard Development.

Stage 1: Data Engineering & ETL Pipeline (Data Preparation)

The project began with highly fragmented raw data spread across 22 separate Excel workbooks representing different administrative units and two distinct ecosystems (Forests and Grasslands).

- **Data Extraction & Merging:** A custom Python script (1_data_cleaning.py) was engineered using Pandas to iterate through all 22 raw Excel sheets and consolidate them into a single master dataframe.
- **Data Cleaning:** The raw dataset was audited for data integrity. The pipeline successfully identified and removed 1,705 duplicate entries to prevent statistical bias.
- **Standardization:** Date and timestamp columns were reformatted into standard datetime objects. Missing categorical variables (like Sex and Distance) were filled, and environmental metrics (Temperature/Humidity) were coerced into strict numeric data types for accurate plotting

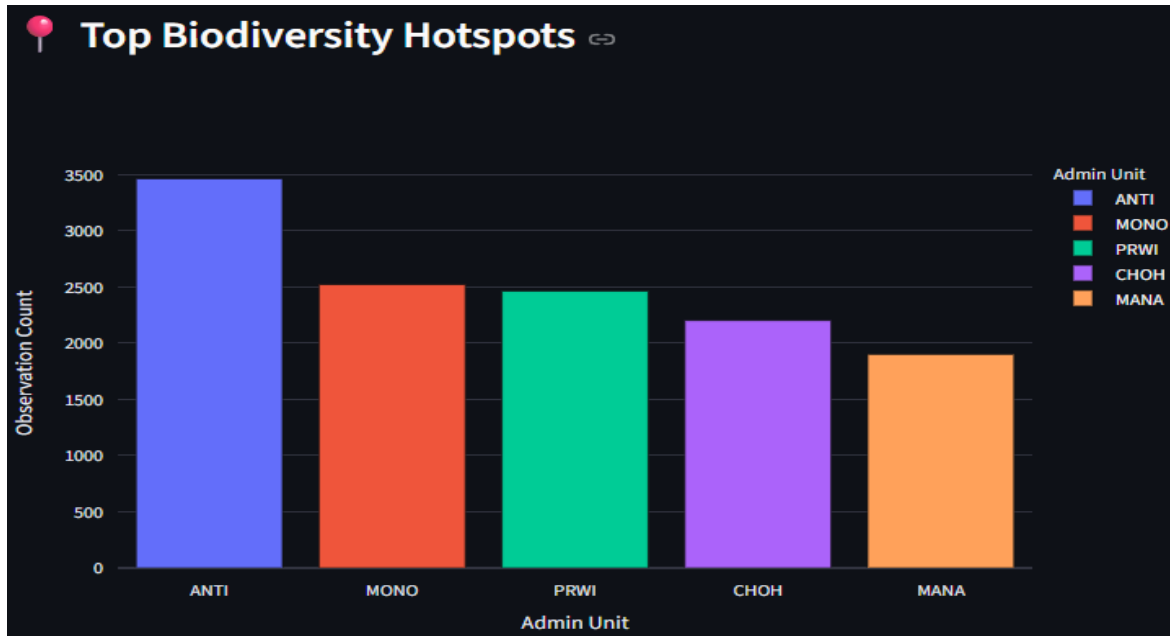
Stage 2: Database Architecture & Data Modeling

To ensure the final dashboard could scale and query data rapidly without loading heavy flat files into memory, the cleaned dataset of 15,372 rows was migrated into a relational database.

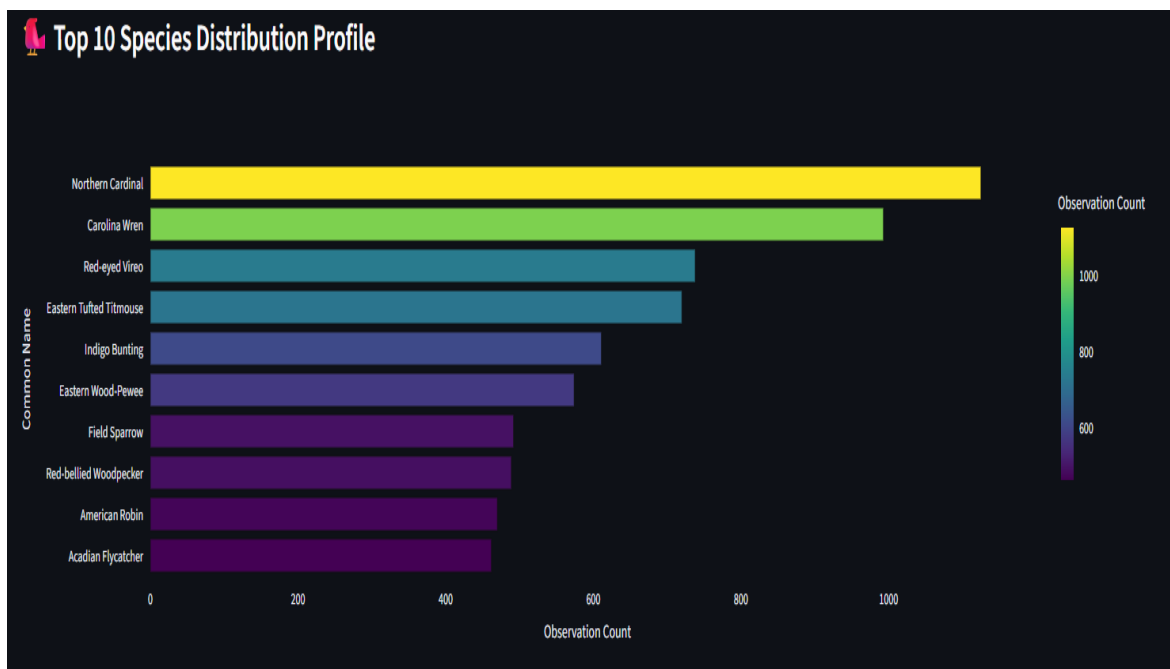
- **SQLite Integration:** The data was exported into a local SQLite database (bird_observations.db). This established a formal relational schema and a centralized bird_data table, ensuring that temporal dates and numeric environmental metrics remain strictly formatted.
- **Dynamic SQL Querying:** The Streamlit application connects directly to this database engine. It utilizes standard SQL SELECT statements to dynamically retrieve only the necessary data payload required for visualization.
- **Performance & Scalability:** To guarantee a fast user experience, the SQL retrieval process utilizes built-in caching, which prevents redundant database calls when users apply dashboard filters. Furthermore, this architecture creates a scalable "single source of truth," meaning future ecological data can be easily appended to the database without needing to rebuild the dashboard

Stage 3: Spatial & Species Analytics (EDA)

1. **Biodiversity Hotspots:** The "Top Biodiversity Hotspots" graph reveals that the ANTI administrative unit is the highest-performing hotspot with nearly 3,500 observations, followed sequentially by MONO (~2,500) and PRWI (~2,400).



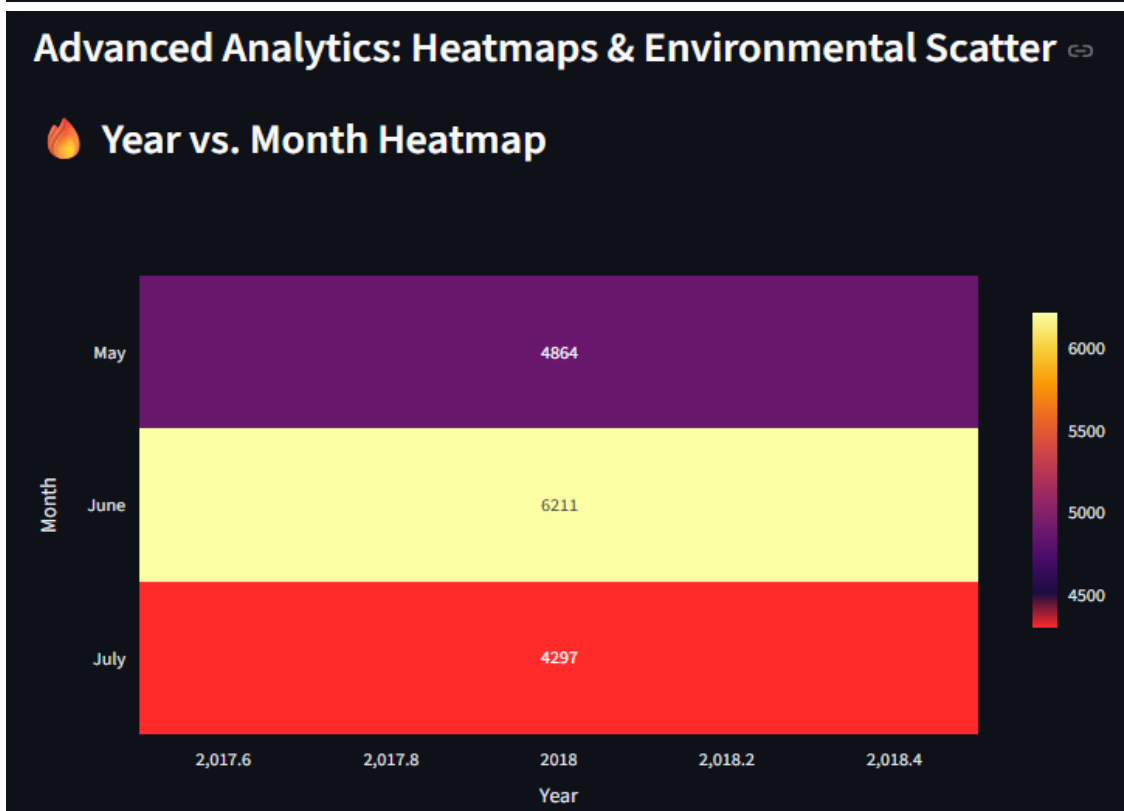
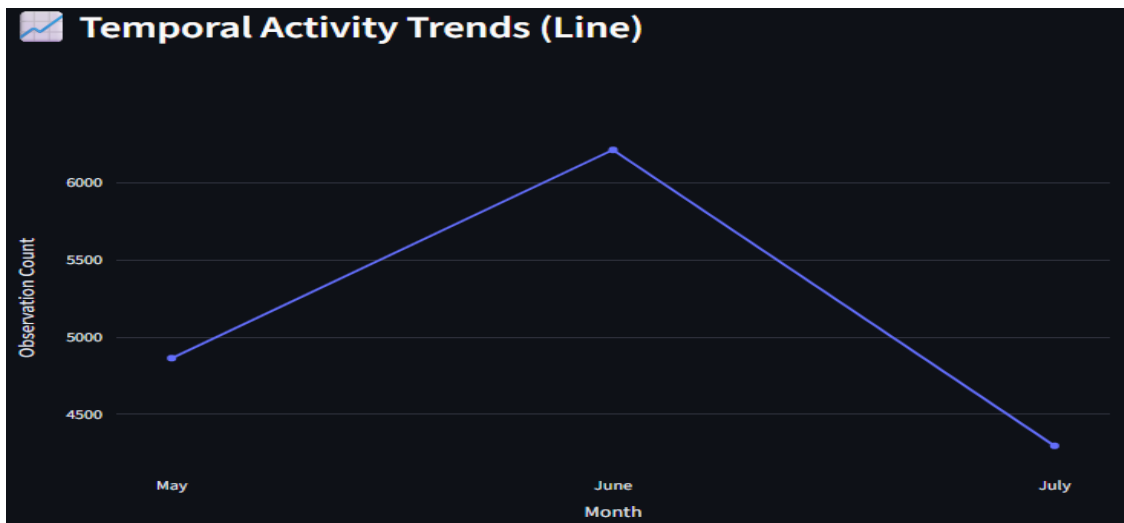
2. **Species Distribution:** According to the "Top 10 Species Distribution Profile," the Northern Cardinal is the most frequently observed species (over 1,000 sightings), closely followed by the Carolina Wren and Red-eyed Vireo.



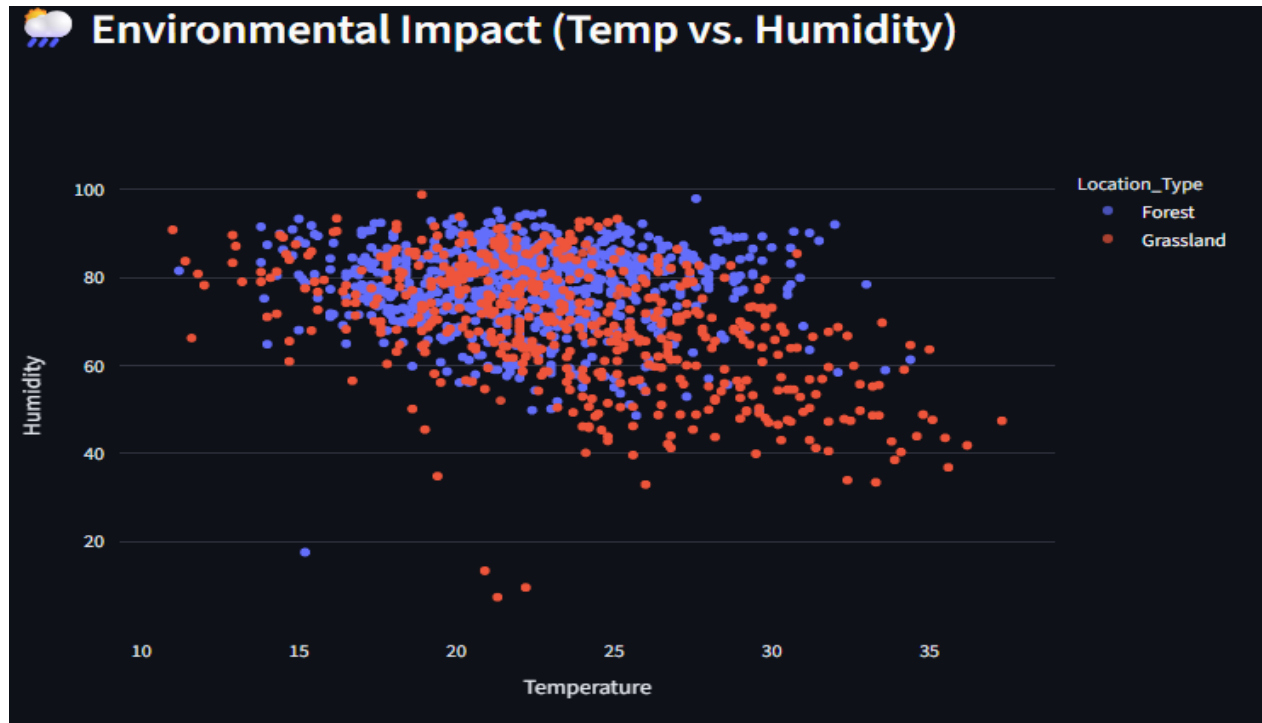
3. **Conservation Focus:** The ecosystem supports 127 unique species, with 8 identified on the Partners in Flight (PIF) Watchlist requiring targeted stewardship and habitat protection.

Stage 4: Temporal Trends & Environmental Impact

1. **Seasonal Peaks:** The "Temporal Activity Trends" line graph and the "Year vs. Month Heatmap" demonstrate a massive spike in avian activity during June (6,211 observations), making it the absolute peak month, bracketed by strong activity in May (4,864) and July (4,297).



2. **Environmental Correlates:** The "Environmental Impact (Temp vs. Humidity)" scatter plot indicates that optimal observation conditions heavily cluster between 15°C and 30°C with humidity ranging from 50% to 90%. Both Forest and Grassland habitats show significant ecological overlap within these weather parameters.



Stage 5: Strategic Recommendations & Dashboard Design

The final deliverable is an interactive web application built with Streamlit and Plotly (3_app.py), allowing stakeholders to filter the database in real-time.

- **Dashboard Layout:** The application utilizes an F-pattern design, organizing the narrative top-to-bottom with headline KPIs positioned in the top-left for immediate visibility. Interactive slicers were placed in a persistent left-hand sidebar so users can cross-filter data by Year, Habitat, and Detection Method.
- **Eco-Tourism Optimization (Actionable Insight):** Because observation volume spikes drastically in mid-summer, tourism marketing campaigns should strictly target the May–July peak window and route bird-watching traffic directly to the highly active ANTI and MONO units.
- **Monitoring Equipment Investment (Actionable Insight):** Data revealed that over 88% of species detection occurs via auditory cues (Singing/Calling), while visual flyovers account for less than 5%. Procurement budgets should be aggressively reallocated away from visual camera traps and heavily invested in passive acoustic monitoring systems.